553/1
BIOLOGY
THEORY
Paper 1
2 Hours

Name : .....

Signature : ..... Personal No : .....

PRE-REGESTRATION EXAMINATIONS 2026

Uganda Certificate of Education

S.4 BIOLOGY THEORY

Paper 1

2 Hours 30 minutes

INSTRUCTIONS TO CANDIDATES

*This paper consists of **two** sections **A**and**B** with seven examination items*

Sectin A has three compulsory items.

*Respond to **all** items in section **A**. All responses in section **A** must be written in the spaces provided.*

*Section **B** consists of **two** parts; Part **I** and Part **II**. Respond to any **one** item from Part **I** and*

*any **one** item from Part **II** and **all** the responses to this sectionmust be written in the answer booklet(s) provided*

Responses to each item in section Bshould start on afresh page.

Item no	Basis code	Weighted score	Score's initian
1	PU		
	PA		
2	CU		
	CA		
3	RU		
	RA		
4/5	EU		

	EA		
6/7	MU		
	MA		
Total weighted score			

SECTION A

Answer all items in this section .All responses to this section must be written in the spaces provided.

ITEM 1

Mr.Omala is a prominate farmer in Mayuge who grows potatoes and sugarcane on large scale. He cares for his crops very well such that can obtain much yields from them ,but in one season at the stage of flowering of his crops , a heavy hail storm striked his garden and different structures on the crops were greatly destroyed. This left Omala worried about the food security his home since potatoes were the staple food in his home. However , after two weeks , he observed new crops appearing and developed may leaves ,branches and roots that enabled them to produce many crop yields.

TASK

a) Identify the processes affected in the plants by the hail storm.

_____

_____

_____

b) Explain how the hail storm affected the normal functioning of the plants

.

Basis code	PU
scores	

c) Explain how the crops were able to grow and produce some yields.

Basis code	PA
scores	

ITEM 2

On his way back from the bar , where the smoke marijuana and take alcohol . Atim encounter a snake ; screamed ,jumped and ran away for his wife until he reached home. On reaching home, his wife asked him what was the problem, but Atim could not talk because of the difficult in breathing and chest pain , he got a wound which took to long to heal due to his condition of insulin deficiency in blood .

TASK

a) Explain the roles of body structures involved in this situation.

b) Explain the physical changes in his body leading to running away up to home

.

Basis code	CU
scores	

How can you advise Atim to manage his healthy situation for the better life.

Basis code	CA
scores	

ITEM 3

Fatuma a 15 year old student is S.3 and Moses a 20 year old student in S.6 were known to be lovers. Fatuma conceived and when Moses learnt of it , he disappeared from the village .Fatuma began encouraging over time as the parents and friends started distancing themselves from her .

TASK

a) What were the potential risks Fatuma is likely to face as a teenager girl?

Basis code	RU
scores	

b) Fatuma of blood group A , delivered a child of blood group O . Moses was eventually arrested and tested to have lood group B but denied the responsibility.

i) Was guilty

ii) May have been innocent

Basis code	RA
scores	

SECTION B

Part 1

Respond to one item from this part. Responses to this section should be written on separate book lets provided.

item 4

A natural forest near a water body was cleared to grow cabbages only on large scale. The cabbages were regularly sprayed with insecticides to control pests, artificial fertlizers applied to boost on the yields but the fishermen complained of reduced fish catch and wild animals are now encroaching on people's crops of the surrounding community reduced harvest of cabbages.

task

a) Explain the cause of the challenges in the community.

b) Explain the measures that should be taken to reduce on the above challenges.

item 5

The residents of Mukono used to dump their domestic wastes in a near by swamp. The fresh vegetation of the swamp together with the dumped waste allowed interaction of different organisms. An investor filled part of the swamp with marrum(red soil) and constructed in factory . The factory directs its wastes into the stream in the swamp .The residents and the entire natural community of organisms no longer enjoy the benefits of the swamp .

task

a) Explain how the investor's actions denied residents and environment the benefits of the swamp they used to get from the swamp.
b) Suggest the importance of conserving such areas described in the scenario.

PART II

Respond to one item from this part. Responses to this part should be written in the answer booklet provides.

item 6

In an attempt to maintain her slender figure. Sarah decides to skip meals of supper and lunch. She mainly depended on to sweets and simple snacks and rarely brushed her teeth. Sarah was smoking marijuana and taking a lot of alcohol but however she became bony ,frequently fall sick , she also had soft bone, toothaches and can not see well at night when she went to the hospital , the doctor's medical report indicated that all these challenges was contributed to her life style and this affected his stay at school.

task

a) Explain how her life style caused the above challenges.
b) Explain how Sarah can over come the above challenges to raise his esteem and stay **at school.**

item 7

David's 8 year old son was attacked by a sickness that resulted in the breathing difficulties ,high body temperature ,vomiting , and soars in the mouth and ulcers in the small intestines .The baby was rushed to the hospital and after some treatment , the breathing normalized and vomiting ceased. The doctors advised that although the son was on treatment but he was anaemic and the most appropriate diet should be given to him to enable him to recover fully in time.

tasks

a) Explain how the sickness affected the normal functioning of the son' s body.
b) Suggest the appropriate diet with reason that will enable the son to recover fully as per the doctor's advise.

END